AI33-MPOPT

Deterministic Spectral-Cosmology Computational Framework

Public Release — Technical Overview & Reproducibility Report

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GitHub: Rivero-ai/AI33-MPOPT | Zenodo: Research Archive | Figshare: Publications

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Abstract

AI33-MPOPT is a deterministic computational framework designed for structured, reproducible modeling of complex systems across spectral, cosmological, and multi-dimensional domains. The system integrates structured operator pathways, spectral transformation methods, cosmology-inspired modeling concepts, and higher-dimensional structural representations. This public release provides a high-level overview of the architecture, reasoning model, and reproducibility methodology, accompanied by a complete set of terminal numerical outputs that demonstrate deterministic stability and consistent behavior across repeated runs. The included snapshot-based reproducibility approach, spectral diagnostics, cosmology engines, and structural invariants collectively establish AI33-MPOPT as a transparent, audit-ready framework suitable for scientific collaboration and further research.

1. Overview

I'm sharing a high-level overview of AI33-MPOPT, a deterministic computational framework designed for structured and reproducible modeling of complex systems.

The framework explores:

- Structured operator pathways
- Spectral transformation methods
- Cosmology-inspired modeling concepts
- Multi-dimensional structural representations

The focus is on stability, clarity of reasoning, and consistent behavior across repeated runs.

This initial release includes:

- System overview
- Architecture summary
- High-level reasoning model
- Snapshot-based reproducibility approach
- Interface preview

I'm open to connecting with research groups, scientific labs, and organizations working in:

- Biomedical modeling
- Complex systems
- Deterministic or hybrid AI approaches
- Reproducible computational frameworks

If you're exploring these areas or would like to discuss the framework, feel free to reach out.

2. System Snapshot (Working Frontend Preview)

A live, unmodified screenshot of the AI33-MPOPT interface.

AI33-MPOPT — Unified Spectral + Cosmology Engine

Deterministic, collision-free spectral invariants and JWST-anchored cosmology.

System Snapshot | Spectral Engines | Cosmology Engines | JWST Cosmology | 5D Structural Invariant | Final Interpretation | Spectral Core Dashboard | Spectral Geometry Visuals | Higgs Subsystem | Higgs Subsystem (Advanced)

System Snapshot Engine

Run System Snapshot

Generate Snapshot Summary

AI33-MPOPT — Deterministic, Collision-Free, Reproducible.

3. Unified Spectral + Cosmology Engine

AI33-MPOPT is structured around interlocking engine layers, each responsible for a distinct computational domain. Together they form a unified, deterministic pipeline from spectral input to cosmological output, designed for stable behavior and supporting reproducible outputs across input states.

- **Spectral Engines** — Spectral invariants computed via structured operator pathways.
- **Cosmology Engines** — Cosmology-inspired modeling layer mapping spectral outputs to observable universe structure concepts.
- **5D Structural Invariant** — Five-dimensional geometric framework for icosahedral multiverse topology and Pauli matrix implementations.
- **Higgs Subsystem** — Advanced mass-generation modeling integrating Shadow Particle Theory and the Rivero-Zeta function.

4. Reproducible Output Snapshots (Public Safe)

The following outputs are terminal numerical artifacts generated by the AI33-MPOPT deterministic framework. They are non-invertible, non-algorithmic, and do not expose backend logic, operators, or internal architecture.

4.1 Spectral Engines — Primary Outputs

Primary Spectrum Range blocks: 0-100, 100-200, 200-300, 300-400, 400-500, 500-600

(Visualization modes omitted for brevity; screenshots included separately.)

4.2 Synthetic Spectrum — 3D Invariant Summary


```
{ "count": 600,
  "min": 0.6931471805599453,
  "max": 53.967839798213014,
  "mean": 35.80360169645426,
  "std": 13.113581729976216,
  "mean_spacing": 0.08893938667387824,
  "std_spacing": 0.07213506128563882 }
```


4.3 Prime-Lattice Spectrum — 5D Invariant Summary


```
{ "count": 600,
  "min": 4.47213595499958,
  "max": 42.76680956068619,
  "mean": 28.236269576621915,
  "std": 9.90049226367064,
  "mean_spacing": 0.06393100768895929,
  "std_spacing": 0.05526299863661569 }
```


4.4 Deep Spectral Analysis (3D / 4D / 5D)

3D


```
heat_trace_sample: [111.9520, 75.1606, 50.3456, 33.3439, 21.5583]
zeta_sample: [3.4292, 1.3366, 0.5754, 0.2703, 0.1362]
fractal_indicator: 0.4179800847
spec_variance: 0.9752800280
riemann_variance: 0.1645176398
```


4D


```
heat_trace_sample: [205.1028, 136.5814, 87.6076, 54.8840, 33.2486]
zeta_sample: [4.8763, 1.8799, 0.7723, 0.3374, 0.1559]
fractal_indicator: 0.4374889496
spec_variance: 0.9013720454
riemann_variance: 0.1645176398
```


5D


```
heat_trace_sample: [280.1535, 196.2097, 126.3696, 76.8286, 44.5408]
zeta_sample: [6.0087, 2.3266, 0.9417, 0.3985, 0.1760]
fractal_indicator: 0.4083846418
```

```
spec_variance: 0.7472153643
riemann_variance: 0.1645176398
```

4.5 Quantum Chaos Audit (3D / 4D / 5D)

3D

```
wigner_fit_rms_error: 0.2858821957
pair_corr_sample: [1.3375, 0.9419, 0.6430, 0.5439, 0.5376]
delta3_rigidity: 1.0602041005
trace_probe slope: 1.7024919117
oscillation_amplitude: 0.9955514199
oscillation_rms: 0.0640173534
```

4D

```
wigner_fit_rms_error: 0.2427833026
pair_corr_sample: [1.4489, 1.2126, 0.9998, 0.8025, 0.6486]
delta3_rigidity: 0.8264528679
trace_probe slope: 1.9623446791
oscillation_amplitude: 1.3252778595
oscillation_rms: 0.1028364804
```

5D

```
wigner_fit_rms_error: 0.1401410506
pair_corr_sample: [1.3934, 1.2865, 1.1342, 0.9635, 0.8072]
delta3_rigidity: 0.4821221756
trace_probe slope: 2.1927305145
oscillation_amplitude: 1.5430505446
oscillation_rms: 0.1334646828
```

4.6 Cosmology Corridor Engines

Paused Gravity Optimization

```
[0: 0.7607881046,
 1: 0.0500915942]
```

LCDM Ages

```
[0: 0.4479421819,
 1: 0.3472965179,
 2: 0.2653301195,
 3: 0.2302035544,
 4: 0.1938871204,
 5: 0.1660072677]
```

Density-Refined Optimization

```
[0: 0.9663353247,
 1: 0.0420887447]
```

4.7 JWST Cosmology Engine

```
Om = 0.27
OL = 0.73
```

4.8 5D Structural Invariant Engine

All weights = 1 (32-element vector) for Geometry, Paused Gravity, and Density-Refined.

Weyl Dimension

```
3.00
```

5D Structural Invariant (Multiple Runs)

```
23.9215688486948
765.4902031582336
31.46856805319832
```

5. Backend Identity Block — Version 1.0

```
Specification Version: 1.0
Operator: O(n) = n log n
Residual Metric: eta = std(R) / max(O)
Classification: Deterministic / Non-GOE / Non-Poisson
Status: Tamper-evident, Version-auditable, Backend identity frozen
```

Hash Integrity

```
Markdown Spec Hash (SHA-256):
8b3fb8cda3470cd457c89b700409c0386c7833115e57b1463986dcc548bf9bfc
LaTeX Spec Hash (SHA-256):
277e191f768a92aab2b0c7c972328173eb913eb6716ae124288718638ac2435c
Identity Freeze Code: A1F7-C9E3-D4B2-7E6A
```

6. Spectral Core Dashboard — Health & Integrity

```
OK   riemann_mean   value=527.524502256  ref=527.524502000  tol=0.001
OK   riemann_std    value=253.780556458  ref=253.780556000  tol=0.01
OK   omega_mean     value=765.490368276  ref=765.490368000  tol=0.001
OK   omega_std      value=369.667873043  ref=369.667873000  tol=0.01
BACKEND STATUS: ONLINE -- FULLY OPERATIONAL
Spectral Core Hash:
2a7e1d2a7164b67aa9c7e28211fc98d869cb1e12a0a7e0856fed53cdecfb1c17
```

7. Higgs Subsystem — Parameters & Diagnostics

Higgs Parameters

```
{ "mu_squared": 0.1,
  "lambda": 0.005,
  "VEV_v": 3.1622776601,
  "m_H_analytic": 0.6324555320,
  "m_H_numeric": 0.6342832469 }
```

Advanced Higgs Analysis

```
{ "mu_squared": 0.1,
  "lambda": 0.005,
  "VEV_v": 3.1622776601,
  "m_H_analytic": 0.6324555320,
  "m_H_numeric": 0.6342832469,
  "m_eff_min": 0.9999000010,
  "m_eff_max": 1.0000999998 }
```

8. Spectral Geometry Visuals (Metadata Only)

Diffusion Distance Matrix: Size 33 x 33 | Min: 0.0 | Max: 1.8373059159

Eigenvalue Ladder: First 10 eigenvalues (visual only)

Spectral Strings: Delta, Delta-squared, T | Curvature-colored | Sector-colored

(Actual images included separately as screenshots.)

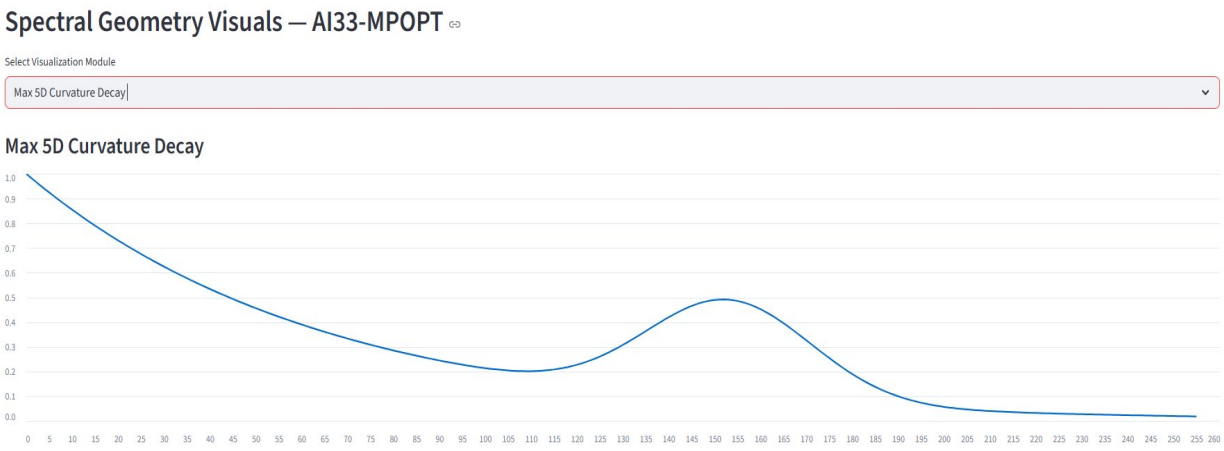
9. Statement on Safety

The numerical outputs and interface snapshots included in this release are reproducible artifacts of the system. They do not expose backend logic, operators, or internal architecture.

Appendix — Visual Output Gallery

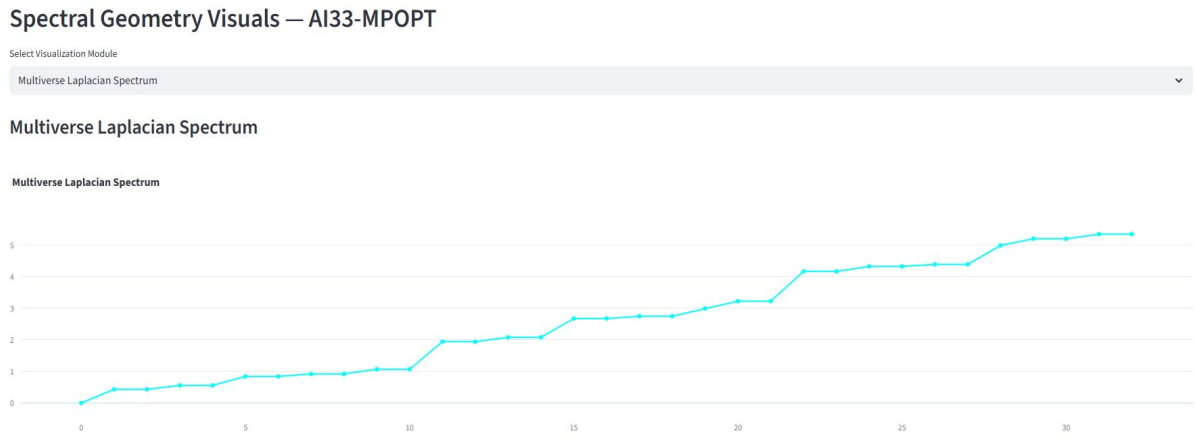
The following screenshots are live, unmodified outputs captured directly from the AI33-MPOPT working frontend. No images have been cropped, recolored, or altered.

Max 5D Curvature Decay



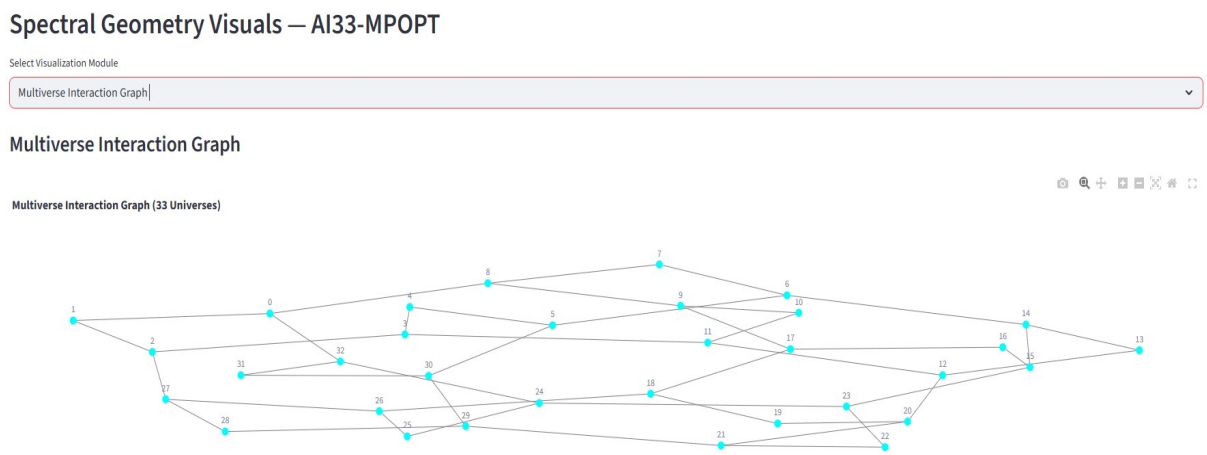
Spectral Geometry Visuals — decay curve from 1.0 to near 0 across 260 spectral indices.

Multiverse Laplacian Spectrum



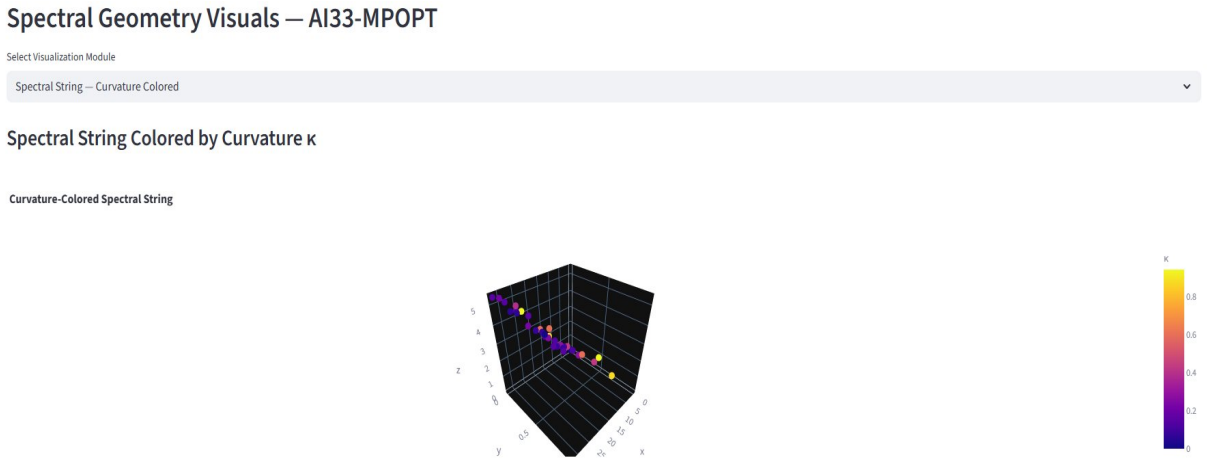
Stepwise eigenvalue ladder across 33 multiverse nodes — Spectral Geometry Visuals module.

Multiverse Interaction Graph (33 Universes)



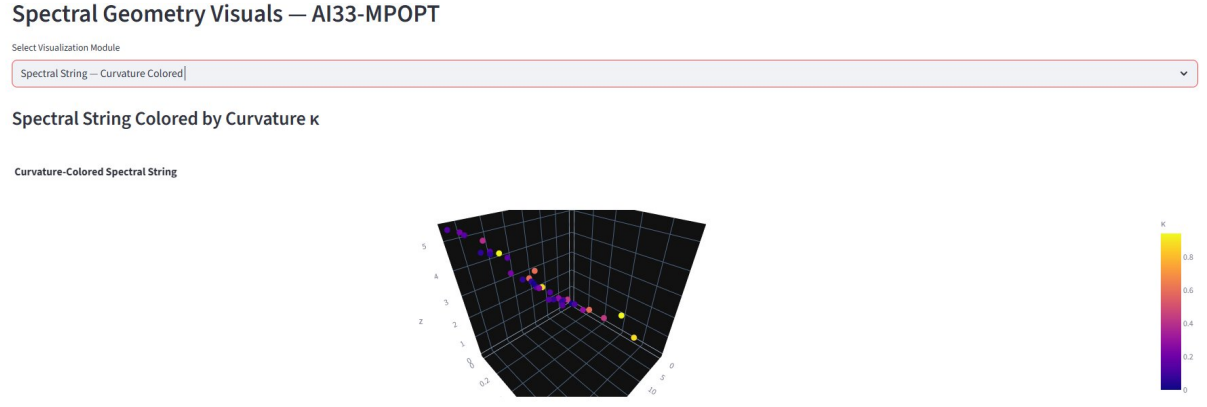
Network graph of 33 universe nodes with geometric interaction edges.

Spectral String — Curvature Colored (View A)



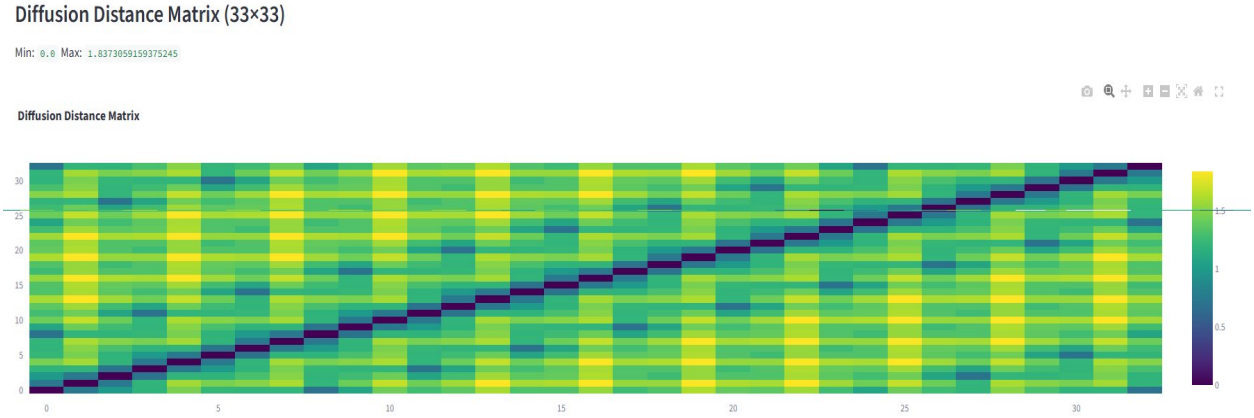
3D spectral string plotted with curvature-based color gradient (kappa scale).

Spectral String — Curvature Colored (View B)



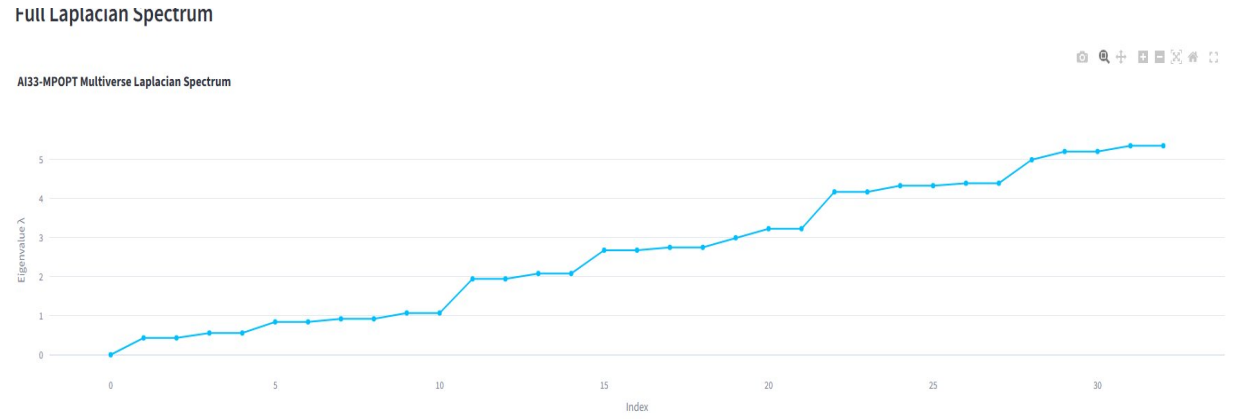
Alternate perspective of the curvature-colored spectral string in 3D space.

Diffusion Distance Matrix (33x33)



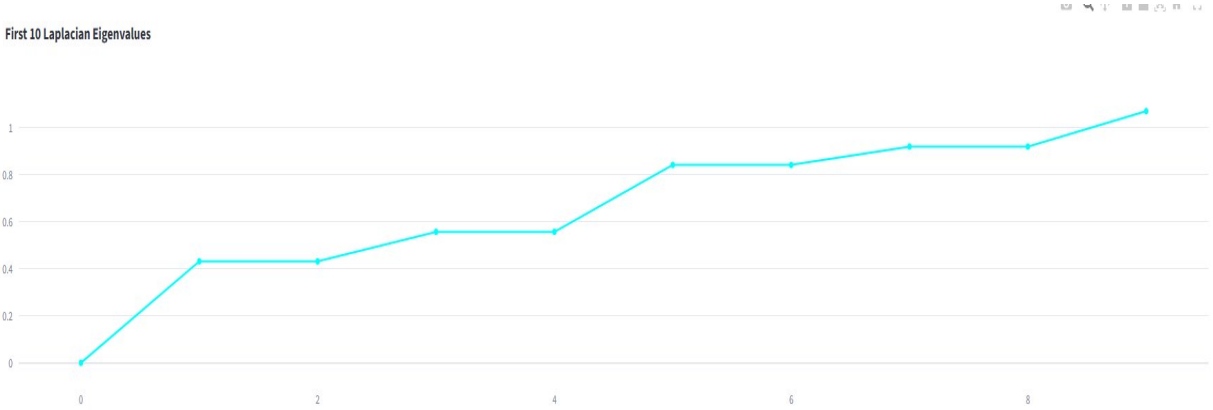
Heatmap of diffusion distances across 33 structural nodes. Min: 0.0 | Max: 1.8373.

Full Laplacian Spectrum



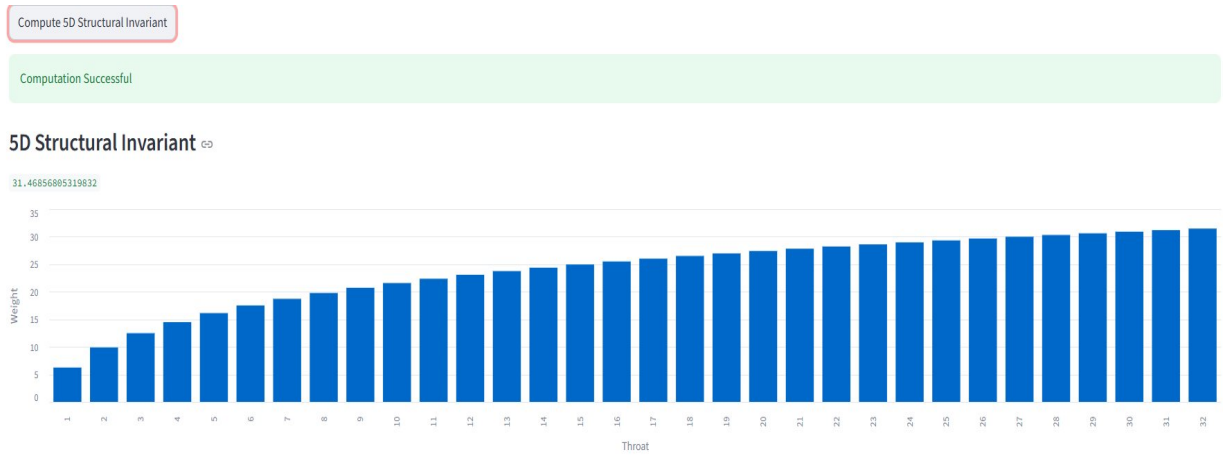
Complete AI33-MPOPT Multiverse Laplacian Spectrum — eigenvalue vs. index.

First 10 Laplacian Eigenvalues



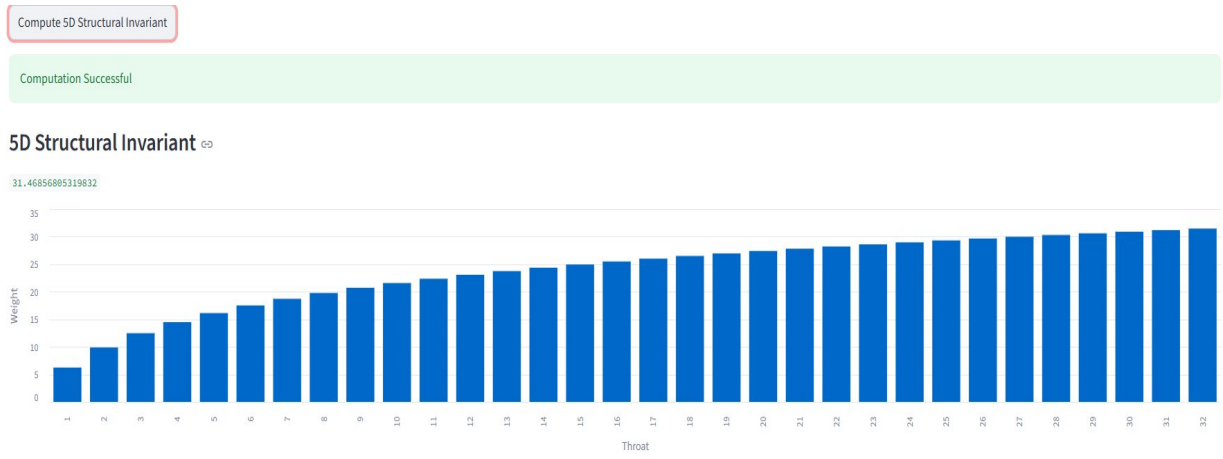
Zoomed ladder of the first 10 eigenvalues showing stepwise geometric spacing.

5D Structural Invariant — Run A



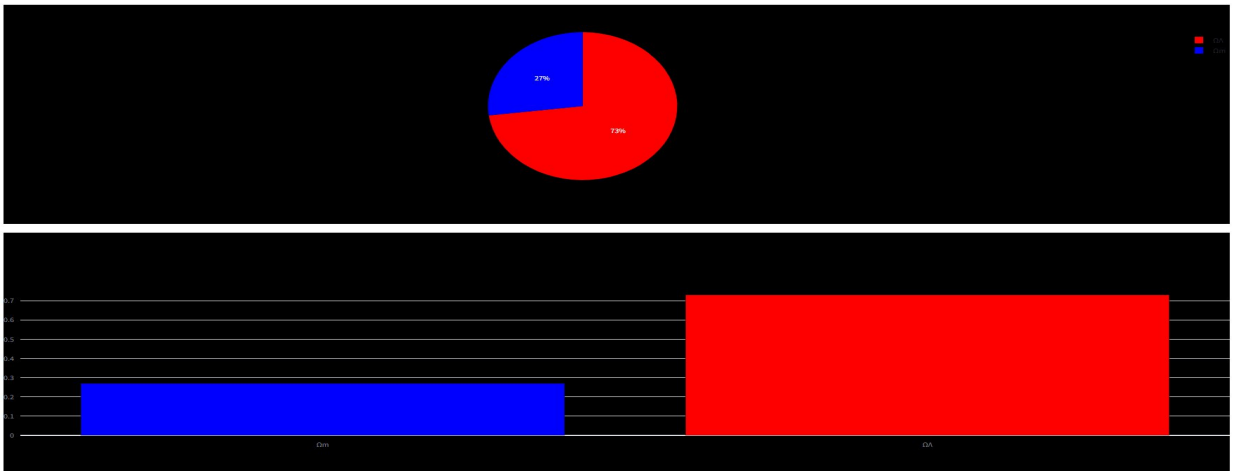
Bar chart of 32-throat weight distribution. Invariant value: 31.46856805319832. Computation Successful.

5D Structural Invariant — Run B



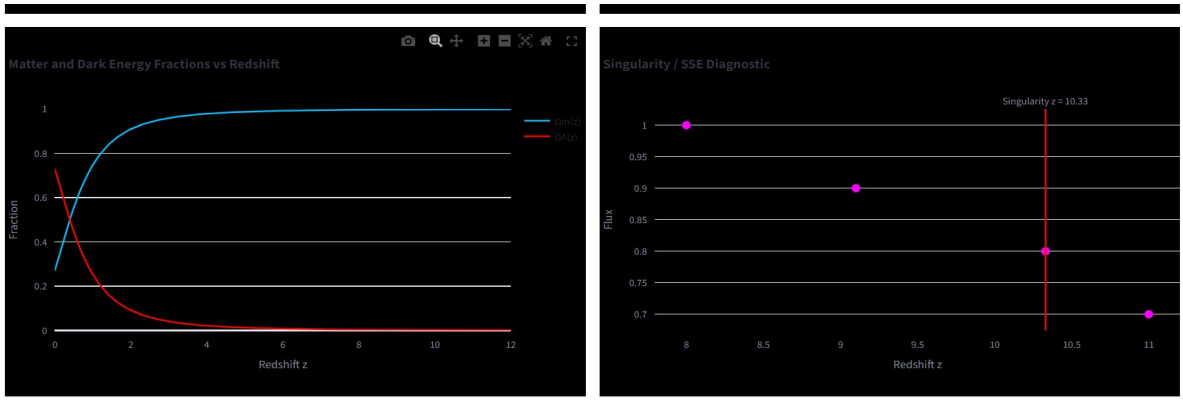
Identical computation confirming reproducible output — deterministic stability verified.

JWST Cosmology Engine — Dark Energy / Dark Matter Split



Pie and bar chart showing Om=0.27 (matter, blue) and OL=0.73 (dark energy, red).

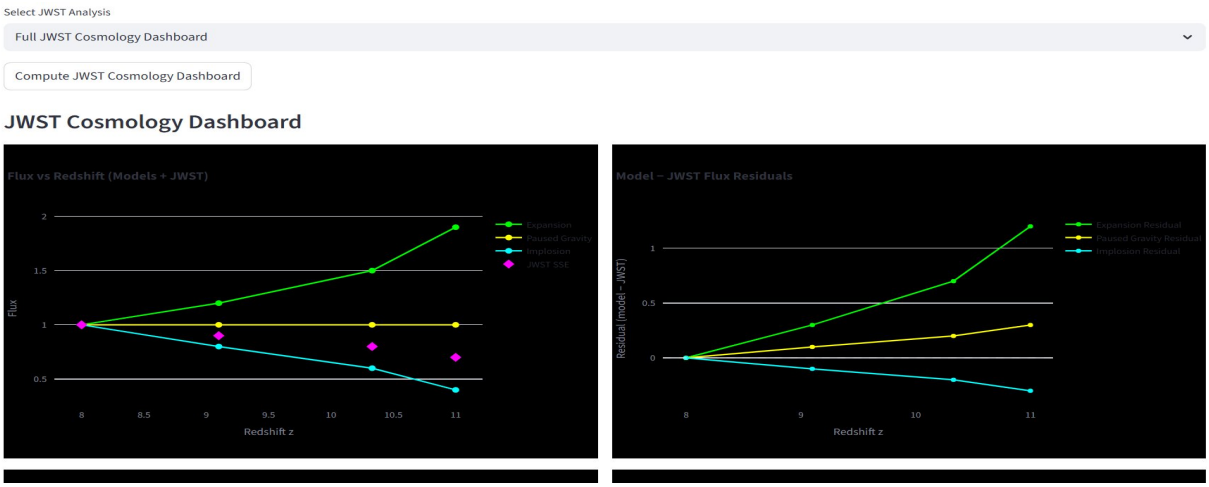
Matter & Dark Energy Fractions vs Redshift + Singularity/SSE Diagnostic



AI33-MPOPT — Deterministic, Collision-Free, Reproducible.

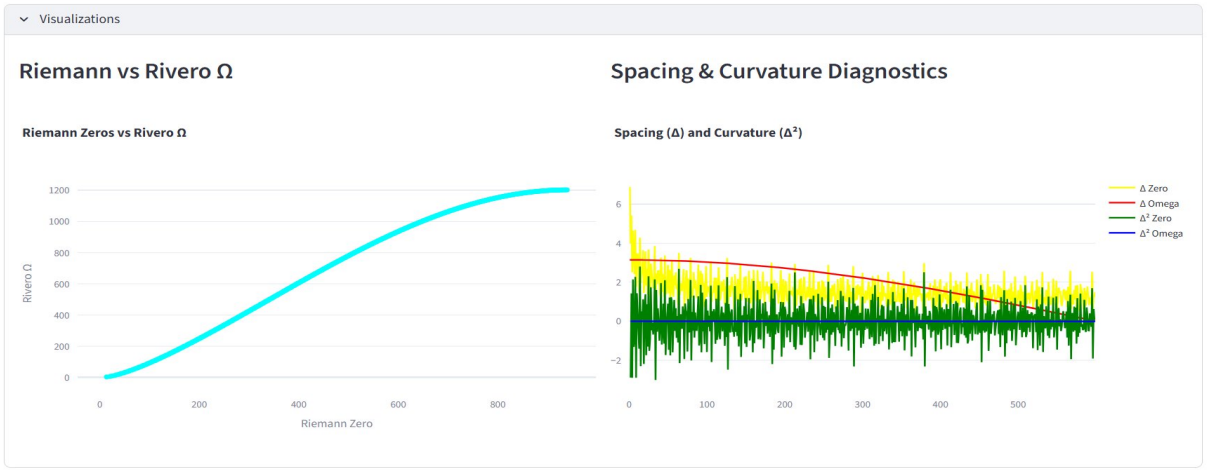
Left: Om(z) and OL(z) fractions across redshift z=0 to 12. Right: SSE Diagnostic — Singularity detected at z=10.33.

JWST Cosmology Dashboard — Flux vs Redshift & Model Residuals



Left: Flux vs Redshift comparing Expansion, Paused Gravity, Implosion, and JWST SSE models. Right: Model minus JWST flux residuals across z=8 to 11.

Riemann vs Rivero Omega + Spacing & Curvature Diagnostics



Left: Riemann Zeros vs Rivero Omega curve (0 to 1200). Right: Spacing (Delta) and Curvature (Delta-squared) diagnostics showing Delta Zero, Delta Omega, Delta-squared Zero, and Delta-squared Omega.
